

Risk Assessment

XBandDigitalModule_20x20 - Risk Assessment

Document Information

Field	Value
Document ID	XBDM-DOC-013
Version	1.0
Date	2026-09-09
Classification	ITAR Restricted

1. Risk Matrix

ID	Description	Probability	Impact	Risk Level	Mitigation	Status
R001	Clock jitter ex	Low	High	Medium	LMX2615-SP (45	Open
R002	JESD204C lane m	Low	Medium	Low	GTY @ 26.5 Gbps	Open
R003	Power sequencin	Medium	High	High	TPS7H3014-SP pr	Open
R004	TID accumulatio	Low	Medium	Low	100+ krad compo	Open
R005	SEL event	Very Low	High	Low	All components	Open
R006	Thermal hotspot	Medium	Medium	Medium	Thermal vias, c	Open
R007	Component obsol	Low	Medium	Low	Multi-source, a	Open
R008	Procurement lea	High	Medium	High	Early procureme	Open
R009	FPGA configurat	Low	High	Medium	Redundant confi	Open
R010	SpaceWire link	Low	Medium	Low	Dual links, err	Open

2. Risk Details

R001: Clock Jitter Exceeds 100 fs

Description: Clock jitter may exceed specification, degrading SNR at 10 GHz input.

Analysis:

Mitigation:

Contingency: If jitter too high, reduce input frequency or use external clock source.

R003: Power Sequencing Violation

Description: Power rails may not sequence correctly, causing latch-up or damage.

Analysis:

Mitigation:

Contingency: If sequencing fails, emergency shutdown via FAULT pin.

R006: Thermal Hotspot

Description: Component temperatures may exceed limits.

Analysis:

Mitigation:

Contingency: Reduce FPGA clock speed or power consumption.

R008: Procurement Lead Time

Description: Long lead times for critical components.

Analysis:

Mitigation:

Contingency: Use rad-tolerant versions (-SEP) with shorter lead times.

3. Risk Tracking

ID	Status	Owner	Target Date	Notes
R001	Open	RF Engineer	2026-10-01	Phase noise mea
R002	Open	FPGA Engineer	2026-10-01	JESD204C simula
R003	Open	Power Engineer	2026-10-01	Sequencer confi
R004	Open	System Engineer	2026-11-01	Radiation test
R005	Open	System Engineer	2026-11-01	SEL test plan
R006	Open	Mechanical Engi	2026-10-01	Thermal simulat
R007	Open	System Engineer	2026-12-01	Component track
R008	Open	Procurement	2026-09-15	Order placement
R009	Open	FPGA Engineer	2026-10-01	Configuration b
R010	Open	FPGA Engineer	2026-10-01	Link redundancy

4. Risk Review Schedule

Review	Date	Focus
PDR	2026-09-15	Architecture, c
CDR	2026-10-15	Schematic, PCB
TRR	2026-12-01	Test readiness
FRR	2027-01-15	Flight readines

5. Revision History

Version	Date	Author	Description
1.0	2026-09-09	System Engineer	Initial release
